The documentation and process conversion measures necessary to comply with this revision shall be completed by 30 June 2003.

INCH-POUND

MIL-PRF-19500/429H
30 December 2002
SUPERSEDING
MIL-PRF-19500/429G
18 October 2001

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, FAST RECOVERY
TYPES 1N5615, 1N5617, 1N5619, 1N5621, 1N5623, 1N5615UL, 1N5617UL, 1N5619UL, 1N5621UL, 1N5623UL, 1N5615US, 1N5617US, 1N5619US, 1N5621US, 1N5623US,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, fast recovery power rectifier diodes. Four levels of product assurance are provided for each encapsulated device as specified in MIL-PRF-19500. Two levels of product assurance are provided each unencapsulated device type.

1.2 Physical dimensions. See figure 1 (similar to D0-7), figures 2 and 3 surface mount (D-5A), and figures 4, and 5 (JANHC and JANKC).

1.3 Maximum ratings.

Col. 1	Col. 2	Col. 3	Col. 4		Col. 5	* Col. 6	Col. 7	Col. 8	Col. 9	Col. 10
Types (1)	V_R	V_{RWM}	I_O		t_{rr}	T_{STG} T_J	I_{FSM} T_A = +100°C I_O = 750 mA dc t_P = 8.3 ms	Barometric pressure (reduced)	$R_{\theta JL}$ $R_{\theta JEC}$	$Z_{\theta JX}$
			T_A = +55°C (2) (3)	T_A = +100°C (3) (4)						
	V dc	V(pk)	A dc	mA dc	ns	°C	A(pk)	mmHg	(5)	°C/W
1N5615	200	200	1	750	150	-65	25	N/A		4.5
1N5617	400	400	1	750	150		25	8		4.5
1N5619	600	600	1	750	250	to	25	8		4.5
1N5621	800	800	1	750	300		25	33		4.5
1N5623	1,000	1,000	1	750	500	+175	25	33		4.5

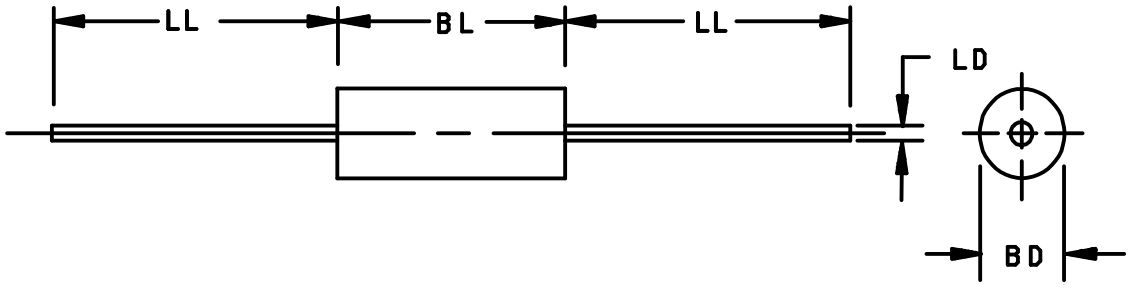
(1) Electrical characteristics for UL and US suffix are identical to the corresponding non-suffix device.
(2) From 1 A dc at T_A = +55°C, to .75 A dc at T_A = +100°C, derate linearly at 5.56 mA/°C.
(3) For the 1 A dc rating at +55°C and the 750 mA dc rating at +100°C, no special mounting, heat sinking, or forced-air flow across exposed areas of the device is required.
*(4) From .75 A dc at T_A = +100°C, to 0 A dc at T_A = +175°C, derate linearly at 10 mA/°C.
(5) $R_{\theta JL} \leq 38$°C/W at L = .375 inch (9.53 mm); for UL and US suffix type devices. $R_{\theta JEC} \leq 7.0$°C/W.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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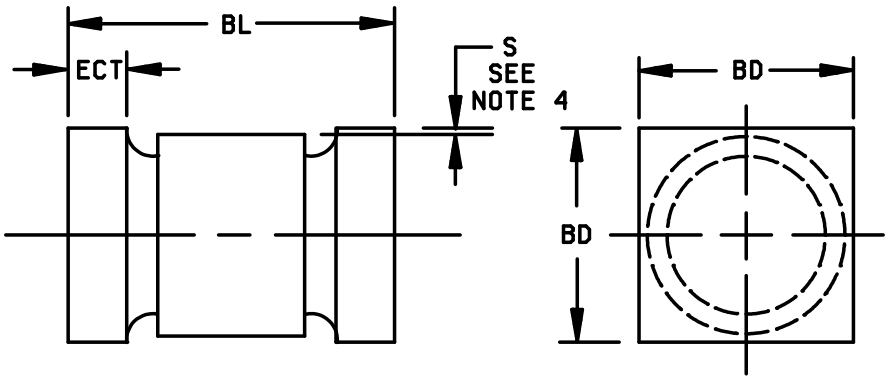


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.065	.110	1.65	2.79	3
LD	.026	.033	.66	.84	
BL	.130	.225	3.30	5.71	4
LL	1.00	1.30	25.40	33.02	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension BD shall be measured at the largest diameter.
4. Dimension BL shall include all uncontrolled areas of the device.

FIGURE 1. Physical dimensions (for non-UL and non-US suffix devices only) (similar to D0-7).

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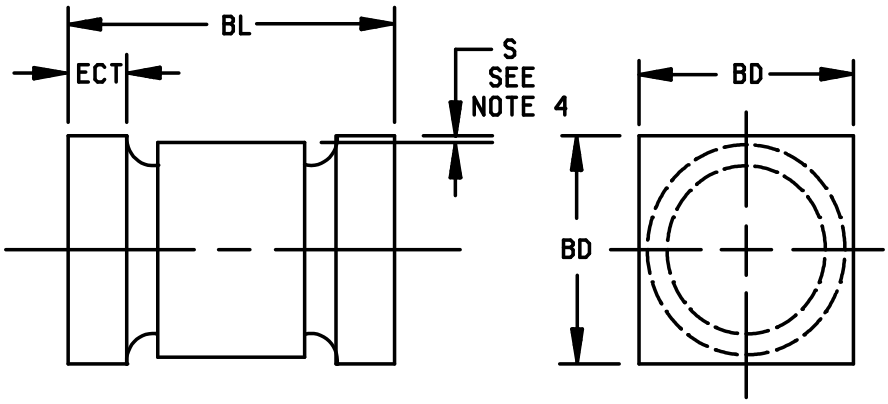


Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.168	.225	4.27	5.72
ECT	.019	.028	0.48	0.71
S	.003		0.08	
BD	.091	.125	2.31	3.18

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions are pre-solder dip.
4. Minimum clearance of glass body to mounting surface on all orientations.

FIGURE 2. Physical dimensions for types 1N5615UL, 1N5617UL, 1N5619UL, 1N5621UL, and 1N5623UL, (surface mount devices) (D-5A).

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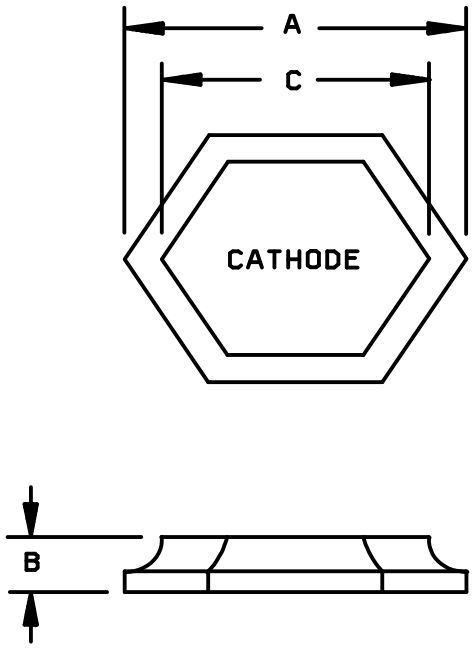


Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.168	.200	4.27	5.08
ECT	.019	.028	0.48	0.71
S	.003		0.08	
BD	.091	.103	2.31	2.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions are pre-solder dip.
4. Minimum clearance of glass body to mounting surface on all orientations.

FIGURE 3. Physical dimensions for types 1N5615US, 1N5617US, 1N5619US, 1N5621US, and 1N5623US, (surface mount devices) (D-5A).

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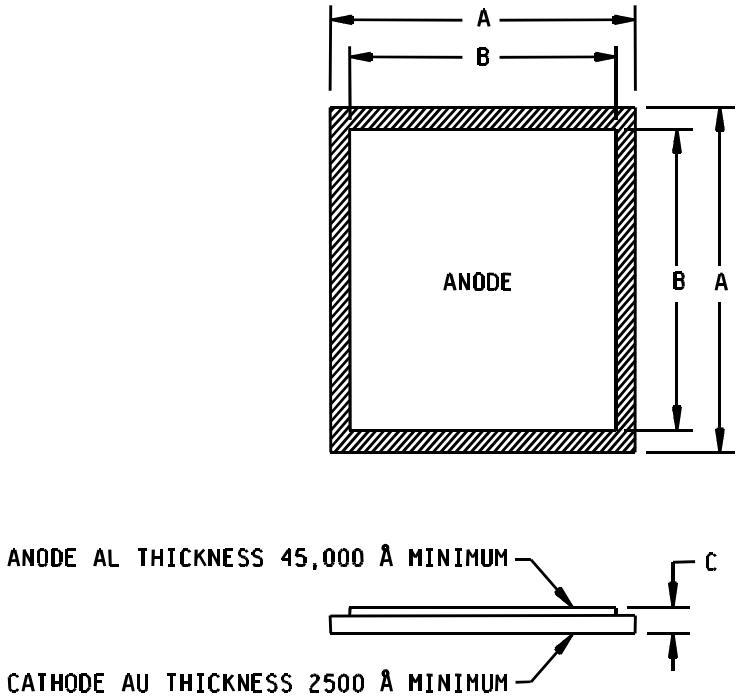
A - version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.047	.053	1.19	1.35
B	.007	.011	0.18	0.28
C	.033	.037	0.84	0.94

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Top metal: Gold 10,000 Å minimum.
   Back metal: Gold 4,000 Å minimum.

FIGURE 4. Physical dimensions, JANHCA and JANKCA die.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.039	.043	1.00	1.09
B	.021	.025	0.53	0.64
C	.008	.012	0.20	0.31

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:
   Top metal: Aluminum (anode) 45,000 Å minimum.
   Back metal: Gold (cathode) 2,500 Å minimum.

FIGURE 5. Physical dimensions, JANHCB and JANKCB die.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

* 3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

EC	End cap.
EG	Epoxy over Glass hermetically sealed EG1 case (see figure 8).
UL	Long-body unleaded or surface mounted diodes (square end-caps).
US	Short-body unleaded or surface mounted diodes (square end-caps).

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (similar to DO-7), figures 2 and 3 surface mount (D-5A), and figures 4 and 5 (JANHC and JANKC).

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* 3.4.1 Lead finish. Unless otherwise specified, lead or end cap finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. When solder alloy is used for finish the maximum lead temperature is limited to 175°C maximum. Where a choice of finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Encapsulant material. In addition to those categories of hermetically sealed package requirements specified in MIL-PRF-19500, fused-metal-oxide to metal shall also be acceptable.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5.1 Marking of UL and US version. For UL and US version only, all marking (see 3.5) may be omitted from the body, but shall be retained on the initial container.

3.6 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for UL and US suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end or a contrasting color band may be used.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E qualification shall be performed herein for qualification or requalification only. In case qualification was awarded to a prior revision of the associated specification that did not request the performance of table II tests, the tests specified in table II herein shall be performed by the first inspection lot to this revision to maintain qualification.

4.2.2 JANHC and JANKC devices. Qualification for die shall be in accordance with MIL-PRF-19500 and as specified herein.

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4.3 Screening (JANS, JANTX, and JANTXV levels). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTXV and JANTX level
1a 1b	Required Required	Not required Required (JANTXV only)
2	Not required	Not required
3a (1) 3c	Required Thermal impedance (see 4.3.3)	Required Thermal impedance (see 4.3.3)
4	Not applicable	Not applicable
5	Not applicable	Not applicable
6	Not applicable	Not applicable
7a 7b	Not applicable Required	Not applicable Required
8	Required	Not required
* 9	Required I_{R1} and V_{FM1}	Not required
10	Method 1038 of MIL-STD-750, condition A	Method 1038 of MIL-STD-750, condition A
11	I_{R1} and V_F, $\Delta I_{R1} \leq 100$ percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.1$ V dc	I_{R1} and V_F
12	Required, see 4.3.1	Required, see 4.3.1
(2) 13	Subgroups 2 and 3 of table I herein: $\Delta I_{R1} \leq 100$ percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.1$ V dc. Scope display evaluation (see 4.5.4).	Subgroup 2 of table I herein: $\Delta I_{R1} \leq 100$ percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.1$ V dc. Scope display evaluation (see 4.5.4).
14a 14b	Not applicable Optional	Not applicable Optional
15	Required	Not required
16	Required	Not required
17	Not required	Not required

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.

(2) $Z_{\theta JX}$ is not required in screen 13, if already previously performed.

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* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1038 of MIL-STD-750, condition B, V_R = full rated V_{RWM} (see 1.3); f = 50-60 Hz, (see 4.5.3 and 4.5.3.1).

- Burn-in set up. Place a single device in your burn-in/life test rack. Adjust power (or current) until the part reaches T_J = 135°C minimum. This is the minimum power/current which must be applied to each device. Use method 3100 to measure T_J. Additional heat generated by self-heating is allowed but shall not cause any device to exceed the maximum temperature rating.

4.3.2 Screening (JANHC and JANKC). Screening of die shall be in accordance with appendix H of MIL-PRF-19500. As a minimum, die shall be 100-percent probed to ensure compliance with table I, subgroup 2. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

4.3.3 Thermal impedance. Thermal impedance $Z_{\theta JX}$ measurements shall be performed in accordance with method 3101 of MIL-STD-750. The maximum limit for $Z_{\theta JX}$ in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control and applied in screening of all subsequent lots. This limit shall not exceed the table I, subgroup 2 limit. The following conditions shall apply:

a. I_H = 5 A minimum.

b. t_H = 10 ms.

c. I_M = 1 mA to 10 mA.

d. t_{MD} = 100 µs maximum.

4.3.3.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3101 of MIL-STD-750 (read and record value $Z_{\theta JX}$) and shall be supplied to the qualifying activity on a 500 piece sample from the qualification lot prior to qualification approval.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$ in table I, subgroup 2 inspection:

a. I_M measuring current - - - - - - - 10 mA.

b. I_H forward heating current - - - - - 10 A.

c. t_H heating time - - - - - - - - - - 10 ms.

d. t_{MD} measurement delay time - - - - - 100 µs maximum.

The maximum limit for $Z_{\theta JX}$ in table I, subgroup 2 is $Z_{\theta JX}$ (maximum) = 4.5°C/W.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. See table III for delta limits when applicable.

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* 4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500. For B5, if a failure occurs, resubmission shall be at the test conditions of the original sample.

	Subgroup	Method	Condition
*	B3	1056	0°C to +100°C, 10 cycles.
*	B3	1051	-55°C to +175°C, 20 cycles.
*	B3	4066	I_{FSM} = rated I_{FSM} (see col. 7 of 1.3); 10 surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 0, V_{RWM} = 0.
	B4	1037	$I_O = I_{O2}$ rated minimum (see col. 4 of 1.3); V_R = rated V_{RWM} (see col. 3 of 1.3, and 4.5.3); 2,000 cycles.
*	B5	1027	$I_O = I_{O2}$ rated minimum (see col. 4 of 1.3); apply V_R = rated V_{RWM} (see col. 3 of 1.3 and 4.5.3) adjust or I_O to achieve T_J minimum; f = 50-60 Hz.
*			Option 1: T_A = + 30°C max. ; T_J = 225°C minimum; t = 216 hours, n = 45, c = 0.
*		or	Option 2: T_A = + 30°C max. ; T_J = 200°C minimum; t = 1,000 hours, n = 45, c = 0.
	B6	3101 or 4081	$R_{\theta JL}$ and $R_{\theta JEC}$ (maximum) (see figure 6).

* 4.4.2.2 Group B inspection, table VIb (JAN, JANTX and JANTXV of MIL-PRF-19500.

	Subgroup	Method	Condition
*	B2	1056	0°C to +100°C, 10 cycles.
*	B2	1051	-55°C to +175°C, 20 cycles.
*	B3	1027	$I_O = I_O$ rated (see col. 4 of 1.3) minimum; adjust I_O to achieve the required T_J apply V_R = rated V_{RWM} (see col. 3 of 1.3), f = 50-60 Hz (see 4.5.3.1).

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. See table III for delta limits when applicable.

* 4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

	Subgroup	Method	Condition
*	B2	1056	0°C to +100°C, 10 cycles.
*	B2	1051	-55°C to +175°C, 20 cycles.
	C2	2036	Tension: Test condition A; weight = 12 pounds; t = 30 seconds. Lead fatigue: Test condition E; weight 1 pounds. NOTE: Both tension and lead fatigue are not applicable for US devices.
*	C6	1027	$I_O = I_O$ rated (rated see col. 4 of 1.3) minimum; adjust I_O to achieve the required T_J apply V_R = rated V_{RWM} (see col. 3 of 1.3), f = 50-60 Hz (see 4.5.3.1).

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4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at an ambient temperature T_A of +25°C ±3°C.

4.5.3 Burn-in and life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees, nor less than 150 degrees.

* 4.5.3.1 Delta T_J rise. Io is adjusted such that the junction rise above ambient shall be delta TJ = 100°C min. for each diode and the junction temperature of each diode is maintained at TJ = 135°C min for burn-in and TJ = 150°C min for life testing. The delta TJ rise is intended to be achieved by device current and not by external or self-heating. (Tj – Ta) $\geq$ 100°C. Deliberate heat sinking or forced air-cooling is prohibited unless otherwise approved by the qualifying activity. Ambient temperature shall be controlled to prevent TJ from exceeding rated TJ. The use of a current limiting or ballast resistor is permitted provided that each DUT still sees the full P_t (minimum) and that the minimum applied voltage, where applicable, is maintained through out the burn-in period.

* 4.5.4 Scope display evaluation. Scope display evaluation shall be stable in accordance with method 4023 of MIL-STD-750. Scope display may be performed on ATE (automatic test equipment) for screening only with the approval of the qualifying activity. Scope display in table I, subgroup 4 shall be performed on a scope. Reverse current (I_{BR}) over the knee shall be 500 µA peak.

* 4.5.6 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3101 or 4081 of MIL-STD-750. Read and record data in accordance with group E herein and shall be included in the qualification report. Forced moving air or draft shall not be permitted across the devices during test. The maximum limit for $R_{\theta JL}$ under these test condition shall be $R_{\theta JL} \leq 4.5$°C/W for L = .375 (9.53 mm).

a. I_H = 5 A dc minimum.

b. $t_{H =}$ thermal equilibrium.

c. I_M = 1.0 mA to 10 mA.

d. t_{MD} = 100 µs maximum.

The device shall be allowed to reach equilibrium at current I_H before the measurement shall be made ($t_H \geq 25$ sec).

LS = Lead spacing = .375 inch (9.53) mm minimum for leaded devices and LS = 0 minimum for unleaded devices as defined (see figure 7) (Metric equivalents are given for general information only):

* 4.5.6 Peak reverse power test. A 20 microsecond half-sine waveform of current shall be used and peak reverse power shall be determined by the product of peak reverse voltage and peak reverse current. A 20 microsecond square waveform may also be used with the approval of the qualifying activity.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
	2071					
Subgroup 1 Visual and mechanical inspection						
Subgroup 2						
Forward voltage	4011	I_F = 3 A dc (pulsed); (see 4.5.1) t_p = 300 µs; 2 percent maximum duty cycle	V_F	0.8	1.6	V_{pk}
Reverse current leakage	4016	DC method	I_{R1}			
1N5615, 1N5615UL, US		V_R = 200 V dc			0.5	µA dc
1N5617, 1N5617UL, US		V_R = 400 V dc			0.5	µA dc
1N5619, 1N5619UL, US		V_R = 600 V dc			0.5	µA dc
1N5621, 1N5621UL, US		V_R = 800 V dc			0.5	µA dc
1N5623, 1N5623UL, US		V_R = 1,000 V dc			0.5	µA dc
Breakdown voltage	4021	I_R = 50 µA dc	$V_{(BR)R}$			
1N5615, 1N5615UL, US				220		V dc
1N5617, 1N5617UL, US				440		V dc
1N5619, 1N5619UL, US				660		V dc
1N5621, 1N5621UL, US				880		V dc
1N5623, 1N5623UL, US				1100		V dc
Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		4.5	°C/W
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	DC method	I_{R2}			
1N5615, 1N5615UL, US		V_R = 200 V dc			25	µA dc
1N5617, 1N5617UL, US		V_R = 400 V dc			25	µA dc
1N5619, 1N5619UL, US		V_R = 600 V dc			25	µA dc
1N5621, 1N5621UL, US		V_R = 800 V dc			25	µA dc
1N5623, 1N5623UL, US		V_R = 1,000 V dc			25	µA dc

See footnote at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Reverse recovery time	4031	Condition B1	t_{rr}			
1N5615, 1N5615UL, US					150	ns
1N5617, 1N5617UL, US					150	ns
1N5619, 1N5619UL, US					250	ns
1N5621, 1N5621UL, US					300	ns
1N5623, 1N5623UL, US					500	ns
Capacitance	4001	V_R = 12 V dc, 0.1 < f < 1.0 MHz	C			
1N5615, 1N5615UL, US					45	pF
1N5617, 1N5617UL, US					35	pF
1N5619, 1N5619UL, US					25	pF
1N5621, 1N5621UL, US					20	pF
1N5623, 1N5623UL, US					15	pF
Scope display	4023	See 4.5.4 n = 116, c = 0				
Subgroup 5						
Not applicable						
Subgroup 6						
Forward surge	4066	I_{FSM} = 25 A (pk); 10 surges of 8.3 ms each at 1 minute intervals, I_O = 750 mA dc; V_R = rated V_{RWM} (see col. 3 of 1.3); T_A = +100°C; (T_{EC} = +100°C for surface mount devices (UL and US version)).				
* Electrical measurement		See table I, subgroup 2 except thermal impedance.				
Subgroup 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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* TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
* Subgroup 1			45 devices c = 0
Thermal shock	1056	-65°C to +175°C, 500 cycles.	
Temperature cycling	1051	500 cycles, -65°C to +175°C.	
Electrical measurements 1/		See table I, subgroup 2	
Subgroup 2			22 devices c = 0
Steady-state dc blocking life	1038	1,000 hours, condition A $V_R = V_{RWM}$ (see col. 3 of 1.3)	
Electrical measurements 1/		See table I, subgroup 2	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices c = 0
Thermal resistance	3101 or 4081	See 4.5.5; $R_{\theta JL} \leq 38$°C/W at L = .375 inch (9.53 mm); for UL and US suffix type devices. $R_{\theta JEC} = 7.0$°C/W.	
Subgroup 5			22 devices c = 0
Barometric pressure, reduced (altitude operation)	1001	Pressure 1N5617, 1N5619 = 8 mm Hg (100,000 ft); 1N5621, 1N5623 = 33 mm Hg (70,000 ft); $V_R = V_{RWM}$ (see col. 3 of 1.3); t = 1 minute (minimum).	
* Subgroup 6			22 devices c = 0
Forward surge	4066	Condition A, I_{FSM} = rated (see col. 5 of 1.3.); 10 surges of 8.3 ms each at 1 minute intervals superimposed on $I_O = 0$; $V_{RWM} = 0$; $T_A = +25$°C.	
Electrical measurement 1/		See table I, subgroup 2.	
* Subgroup 7			
Peak reverse power		See 4.5.5 and figure 5 herein. Peak reverse power (P_{RM})= shall be characterized by the supplier and this data shall be available to the Government. Test shall be performed on each sublot.	
* Electrical measurement		During the P_{RM} test, the voltage (V_{BR}) shall be monitored to verify it has not collapsed. Any collapse in V_{BR} during or after the P_{RM} test or rise in leakage current (I_R) after the test that exceeds I_{R1} in Group A shall be considered a failure to that level of applied P_{RM}. Progressively higher levels of P_{RM} shall be applied until failure occurs on all devices within the chosen sample size to characterize each sublot.	

* 1/ The sample for this step stress requirement shall be determined by the supplier. A statistical significant sample size is required.

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TABLE III. Groups A, B, and C electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Reverse current leaking change	4016	DC method	ΔI_{R1} 4/		For JAN, JANTX, and JANTXV, ≤ 250 nA dc or 100 percent, whichever is greater; for JANS, ≤ 100 nA dc or 100 percent, whichever is greater.	
2.	Forward voltage change	4011	I_F = 3 A dc; Pulsed (see 4.5.1) t_p = 300 µs; 2 percent maximum duty cycle.	ΔV_{F1} 4/		± 50 mV dc maximum change from previous measured value.	

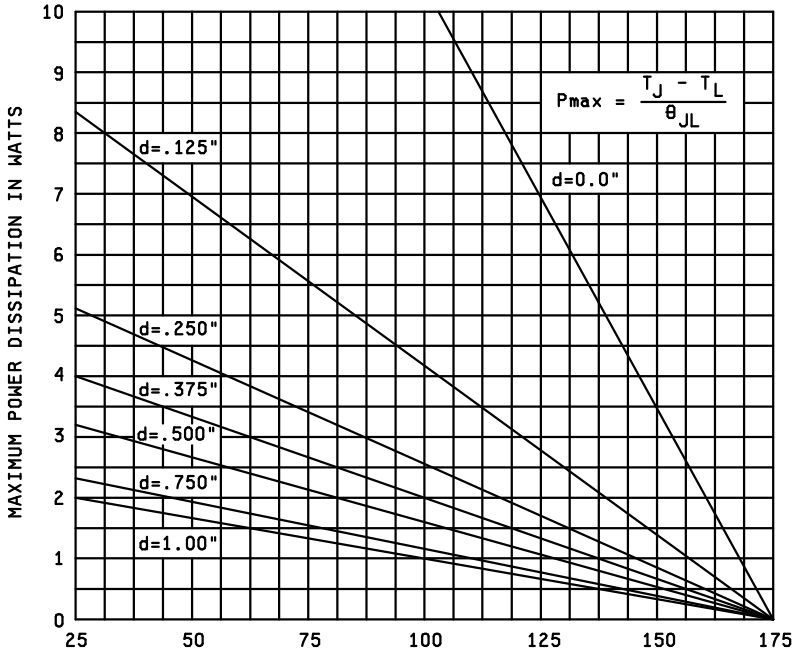
1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table III herein, step 2.
   b. Subgroup 4, see table III herein, step 2.
   c. Subgroup 5, see table III herein, step 1.

2/ The electrical measurements for table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table III herein, step 1.
   b. Subgroup 6, see table III herein, step 1.

3/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table III herein, step 1 (JANS).
   b. Subgroup 6, see table III herein, step 1 and 2 (JANS), step 1 (JAN, JANTX, and JANTXV).

4/ Devices which exceed the group A limits for this test shall not be accepted.

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Maximum lead temperature in °C (T_L) at point d from body (see table below).
For maximum operating junction temperature of +175°C with equal two-lead conditions.

d (see notes 1 and 2)	$R_{\theta JL}$
Dimensions	°C/W
Inches mm	
.000 (0.00)	7
.125 (3.18)	18
.250 (6.35)	29
.375 (9.53)	38
.500 (12.70)	45
.750 (19.05)	62
1.000 (25.40)	76

NOTES:
1. Lead distance point d and thermal resistance data.
2. Thermal resistance test performed at .375 inch (9.53 mm).
3. $R_{\theta JEC} \leq 7$°C/W.

FIGURE 6. Maximum power versus lead temperature.

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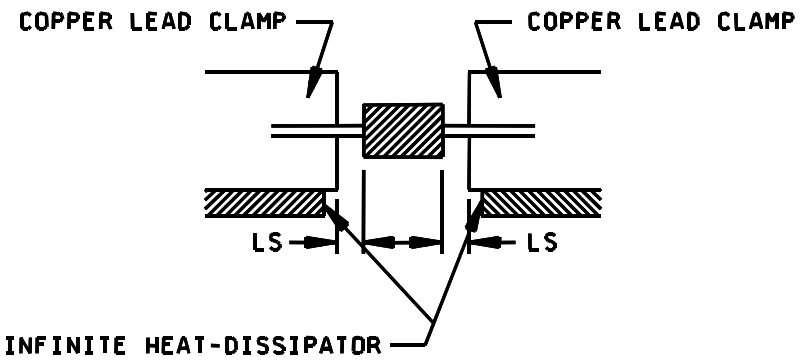
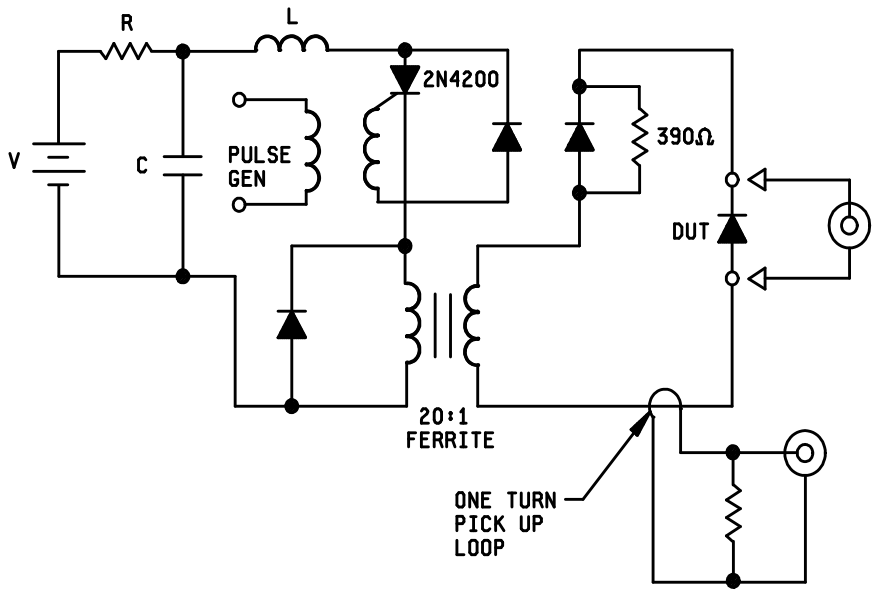
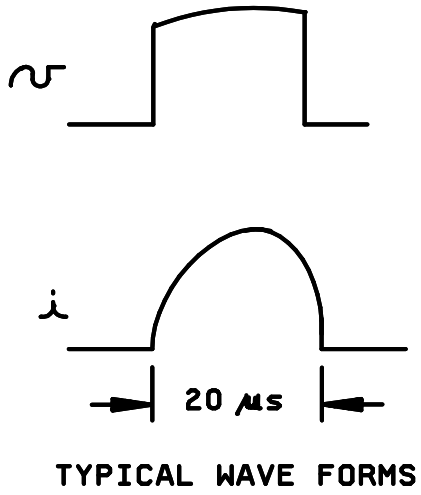


FIGURE 7. Mounting arrangement.

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NOTES:
1. L - 13T #22 pm 1 inch (25.4 mm) diameter form (air core).
2. C - 1 to 10 µfd to give a 20 µs pulse width.
3. V - adjustable to 200 volts for power desired in D.U.T.



TYPICAL WAVE FORMS

FIGURE 8. Peak reverse power measurement circuit and waveform.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.2.1).

c. The lead finish as specified (see 3.4.1).

d. Type designation and quality assurance level.

e. Packaging requirements (see 5.1).

f. For die acquisition, the JANHC or JANKC letter version shall be specified (see figures 4 and 5).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer’s List (QML) No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.4 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N5615) will be identified on the Qualified Manufacturers List (QML).

JANHC and JANKC ordering information		
PIN	Manufacturer	
	14552	33178
1N5615	JANHCA1N5615 JANKCA1N5615	JANHCB1N5615 JANKCB1N5615
1N5617	JANHCA1N5617 JANKCA1N5617	JANHCB1N5617 JANKCB1N5617
1N5619	JANHCA1N5619 JANKCA1N5619	JANHCB1N5619 JANKCB1N5619
1N5621	JANHCA1N5621 JANKCA1N5621	JANHCB1N5621 JANKCB1N5621
1N5623	JANHCA1N5623 JANKCA1N5623	JANHCB1N5623 JANKCB1N5623

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6.5 Changes from previous issue. The margins of this revision are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 11
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Review activities:
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Preparing activity:
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3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, FAST RECOVERY TYPES 1N5615, 1N5617, 1N5619, 1N5621, 1N5623, 1N5615UL, 1N5617UL, 1N5619UL, 1N5621UL, 1N5623UL, 1N5615US, 1N5617US, 1N5619US, 1N5621US, 1N5623US, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan.barone@dscc.dla.mil
c. ADDRESS Defense Supply Center, Columbus ATTN: DSCC-VAC, P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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